

Common Viva Questions Asked in Design and Analysis of Algorithms Research Paper Presentation

1. What is the main objective of your research paper?

Answer: The main objective of my research paper is to analyze and compare the efficiency of algorithms for a specific problem and to propose an optimized approach with better time and space complexity.

2. Which algorithm have you used in your paper and why?

Answer: I have used [mention your algorithm name, for example Merge Sort, Dijkstra, or Dynamic Programming]. I used it because it provides optimal results with less time complexity compared to other approaches for this problem statement.

3. What is the time complexity and space complexity of your algorithm?

Answer: The time complexity of my algorithm is $O(n \log n)$ in the average case and $O(n)$ or $O(n^2)$ in the worst case depending on the algorithm. The space complexity is $O(n)$. I have analyzed it using asymptotic notations.

4. What methodology did you follow for your research?

Answer: I followed standard research methodology which includes problem identification, literature survey, algorithm design, implementation, testing with sample data and complexity analysis.

5. What is the difference between your work and existing work?

Answer: Existing works focus more on theoretical analysis. My work provides practical implementation, comparative analysis and performance improvement in terms of execution time and resource utilization.

6. What are the real world applications of your research?

Answer: My research can be applied in areas like data searching and sorting, network routing, optimization problems, artificial intelligence and large scale data processing where efficient algorithms are required.

7. What tools and languages did you use for implementation?

Answer: I have used Python language for implementation and analysis because it is efficient for algorithm testing. I also used research papers from IEEE and Google Scholar for reference.

8. What are the limitations of your proposed approach?

Answer: The limitation of my approach is that its performance may decrease when the input size is extremely large and it may require more memory. This can be improved in future work.

9. What is the future scope of your research?

Answer: In future this work can be extended by using parallel algorithms, machine learning optimization and testing on larger datasets to further improve efficiency.

10. What did you learn from this research paper work?

Answer: From this research I learned how to design an efficient algorithm, analyze its complexity, compare different approaches and how to write and present a research paper systematically.